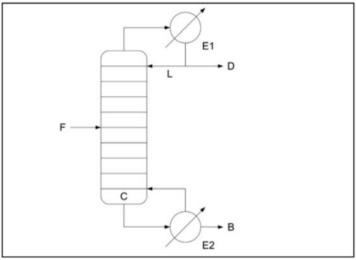

Figure 3: Very simplified schema for a generic distillation process

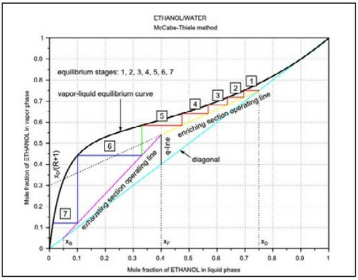

Figure 4: Distillation of the ethanol/water mixture

```
Vertical lines
    end
    i=i+1;
end

// Exhausting section
SS=(yi-xB)/(xi-xB);
yp(i)=SS*(xp(i)-xB)+xB;
y=yp(i);
plot([xp(i),xp(i)],[yp(i-1),yp(i)],"g-"); // Vertical line
while (xp(i)>xB),
    xp(i+1)=fsolve(0.01,equilib); // Steps x coordinates
    yp(i+1)=SS*(xp(i+1)-xB)+xB;   // Steps y coordinates
    y=yp(i+1);
    plot([xp(i),xp(i+1)],[yp(i),yp(i)],"b-"); // Steps plot
    if (xp(i+1)>xB) then
        plot([xp(i+1),xp(i+1)],[yp(i),yp(i+1)],"b-"); //
Vertical lines
    end
    i=i+1;
end
```

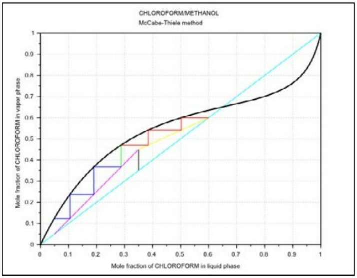

Figure 5: Distillation of the chloroform/methanol mixture

Figures	Notes
1	Reference 3 at the end of the article does not give the data beyond the azeotrope.
3	F, feed; L, liquid; D, distillate; R=L/D, reflux ratio; B, bottom product; C, distillation column; E1, condenser; E2, reboiler.
4	xF (feed composition)=0.4; xD (distillate composition)=0.75; xB (bottom product composition)=0.05; R=1.5; q=1.
5	xF=0.35; xD=0.6; xB=0.05; R=1.5; q=1.

With Scilab, it's possible to calculate the equation of a vapour-liquid equilibrium curve with the use of a rational function, at least as a first guess. Probably, in the future, it will be necessary to improve this kind of rational function. Then it will be possible to simulate a distillation process to obtain the number of equilibrium stages. I hope this article will stimulate the reader's curiosity about Scilab. END

References

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